

AIRLINE DELAY ANALYTICS REPORT

Comprehensive Flight Performance & Delay Analysis

Dataset: U.S. Domestic Flights | 5.7 Million Records | 14 Airlines

github.com/bhaveschhimwal/airline-delay-analytics

5.71M

Total Flights

17.91%

Delayed Flights

4.41 min

Avg Arrival Delay

14

Airlines Analyzed

Delay Category	Total Flights	Share
Early / On Time	3,627,112	63.5%
Slight Delay (1–30 min)	1,063,398	18.6%
Moderate Delay (31–60 min)	704,406	12.3%
Major Delay (> 60 min)	319,092	5.6%

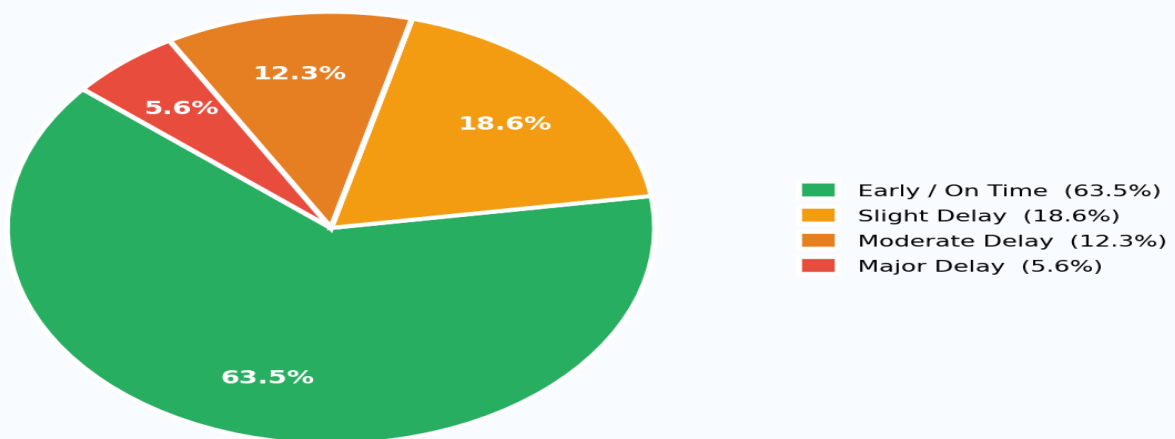
1. Executive Summary

This report presents a comprehensive data-driven analysis of U.S. domestic airline flight delays using a dataset of **5,714,008 flights** across **14 major carriers**. The analysis covers delay patterns by airline, departure time, month, route distance, and airport — providing actionable insights into the key drivers of flight delays.

- ✈ **17.91%** of all flights experienced a delay, with an average arrival delay of **4.41 minutes**.
- ✈ **Spirit Airlines** had the worst on-time performance (+14.47 min avg delay); **Alaska Airlines** actually arrived early on average (−0.98 min).
- ✈ **June** is the worst month for delays (avg +9.60 min); **September & October** are best (avg −0.77 / −0.78 min).
- ✈ **Evening departures** suffer the highest delay rate (24.85%); **Morning flights** are the safest choice (12.22%).
- ✈ **Chicago O'Hare (ORD)** leads major hub delays with an avg departure delay of 14.03 min.

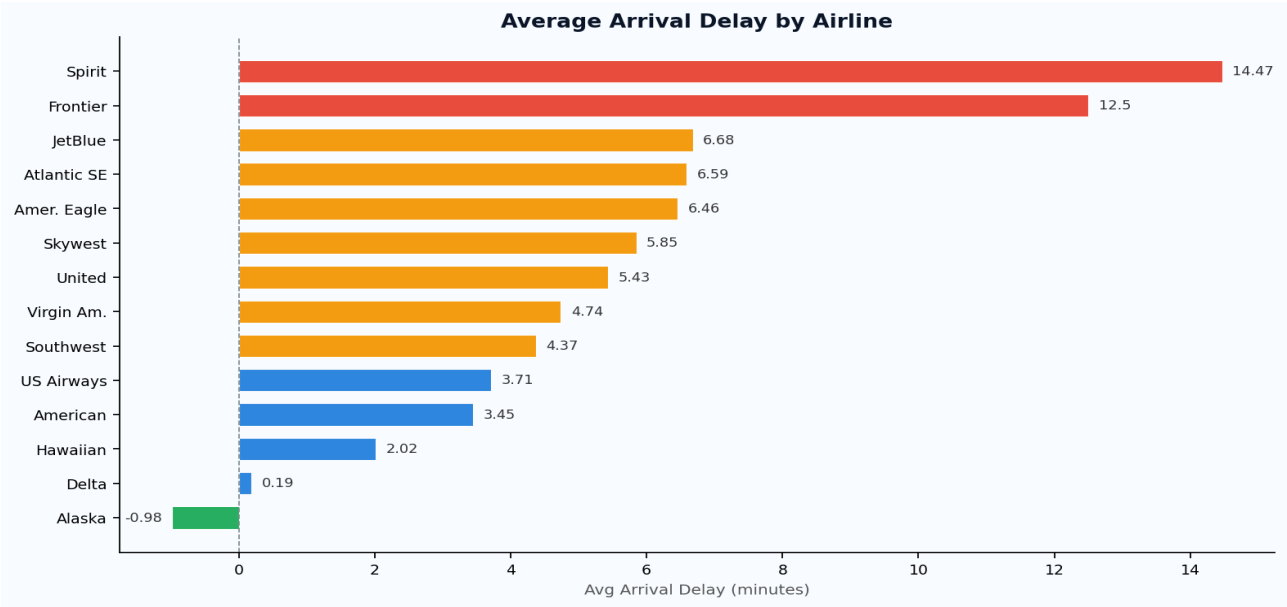
2. Delay Status Distribution

Flight Delay Status Distribution



Nearly **64%** of flights arrived early or on time. Slight delays (1–30 min) account for the bulk of disruptions at 18.6%, while major delays over 60 minutes affect only 5.6% of flights but disproportionately impact passenger experience.

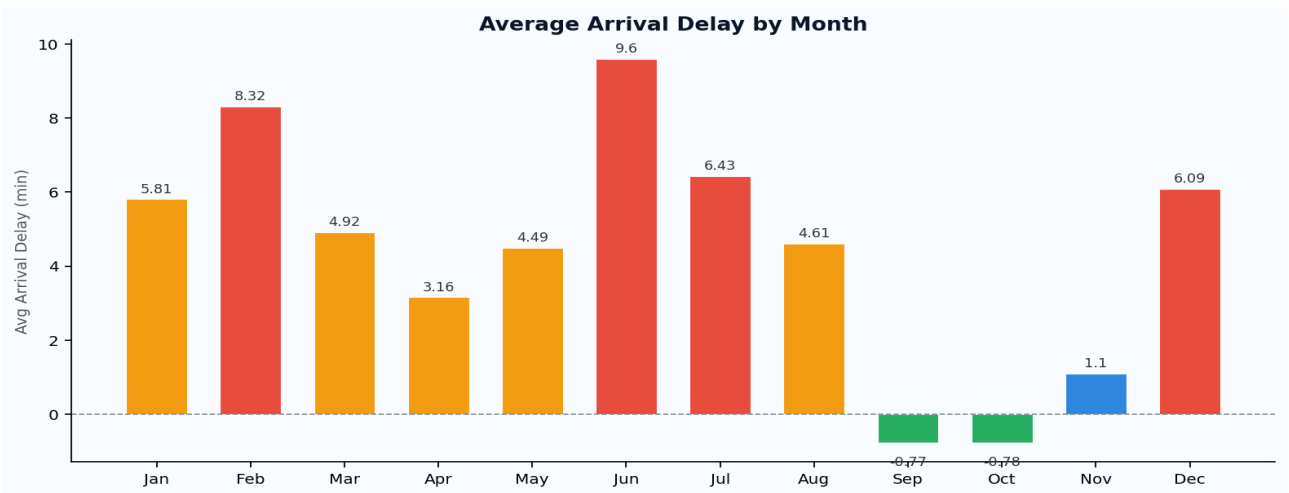
3. Airline Performance Comparison



Rank	Airline	Avg Delay (min)	Performance
1	Spirit Air Lines	14.47	Poor
2	Frontier Airlines	12.50	Poor
3	JetBlue Airways	6.68	Below Avg
4	Atlantic Southeast	6.59	Below Avg
5	American Eagle	6.46	Below Avg
6	Skywest Airlines	5.85	Below Avg
7	United Air Lines	5.43	Below Avg
8	Virgin America	4.74	Average
9	Southwest Airlines	4.37	Average
10	US Airways	3.71	Average
11	American Airlines	3.45	Average
12	Hawaiian Airlines	2.02	Good
13	Delta Air Lines	0.19	Good
14	Alaska Airlines	-0.98	Best

Key Insight: Ultra-low-cost carriers (Spirit, Frontier) suffer the most delays, likely due to tighter turnaround schedules and higher aircraft utilization. Legacy carriers like Delta and Alaska lead on punctuality.

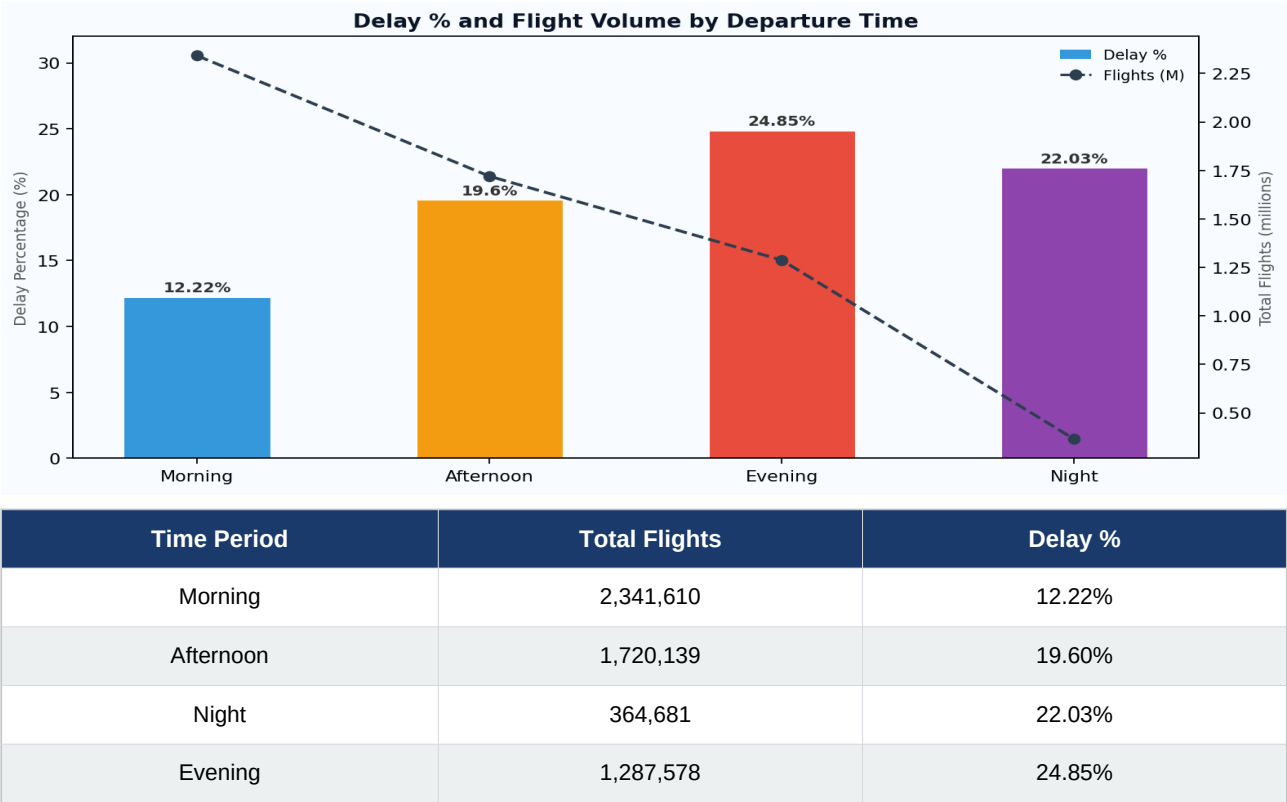
4. Monthly Delay Trends



Month	Total Flights	Avg Delay (min)	Assessment
June	492,847	9.60	Worst
February	407,663	8.32	High
July	514,384	6.43	High
December	469,717	6.09	High
January	457,013	5.81	High
March	492,138	4.92	Moderate
August	503,956	4.61	Moderate
May	489,641	4.49	Moderate
April	479,251	3.16	Low
November	462,367	1.10	Low
September	462,153	-0.77	Best
October	482,878	-0.78	Best

Key Insight: Summer (June–July) and winter holidays (December–February) drive peak delays due to high passenger volumes and adverse weather. Fall months (Sep–Oct) are optimal.

5. Delay by Departure Time Period



Key Insight: Evening departures are nearly **twice as likely** to be delayed compared to morning flights (24.85% vs 12.22%). Delays accumulate across the network throughout the day — a classic delay-propagation effect.

6. Top 10 Airports by Average Departure Delay

#	Airport	Total Flights	Avg Dep. Delay (min)
1	ASE – Aspen/Pitkin County	3,263	17.39
2	EGE – Eagle County Regional	1,198	15.86
3	COU – Columbia Regional	1,241	14.67
4	TTN – Trenton-Mercer	2,771	14.20
5	ORD – Chicago O'Hare Intl	276,554	14.03
6	EWR – Newark Liberty Intl	98,341	13.55
7	BWI – Baltimore/Washington	84,329	13.26
8	LGA – New York LaGuardia	94,834	13.17
9	MDW – Chicago Midway	78,685	12.77
10	IAD – Dulles International	33,664	12.63

Key Insight: Small mountain airports (ASE, EGE) top the list due to weather sensitivity, while NY-area airports and Chicago hubs reflect systemic congestion.

7. Flight Distance vs. Delay

Distance Category	Threshold	Total Flights	Delay %
Medium Distance	500–1,500 miles	2,862,690	18.27%
Long Distance	> 1,500 miles	777,368	17.62%
Short Distance	< 500 miles	2,073,950	17.53%

Key Insight: Delay rates are remarkably consistent across all distance categories (17.5–18.3%), confirming that distance alone is not a primary delay driver.

8. Key Conclusions & Recommendations

Book Morning Flights	Morning departures have the lowest delay rate at 12.22% — nearly half the evening rate. The earlier you fly, the less accumulated network delay affects you.
Choose the Right Airline	Alaska Airlines and Delta consistently outperform the industry. Ultra-low-cost carriers (Spirit, Frontier) should be booked with extra buffer time for connections.
Avoid Peak Travel Months	June is the single worst month (+9.60 min avg). September and October offer the smoothest travel experience, with flights actually arriving early on average.
Beware NY & Chicago Hubs	Itineraries routing through ORD, EWR, LGA, or MDW should include extra connection time or consider alternative routing through less congested airports.
Distance Doesn't Predict Delay	All distance brackets show ~17–18% delay rates. Focus on airline, time of day, and month rather than route length when optimising for punctuality.

9. Technical Notes – SQL Schema

Analysis performed using MySQL on a **flights** table with 5,714,008 rows. Key engineered columns: **is_delayed** (binary), **delay_status** (Early/Slight/Moderate/Major), **time_period** (Morning/Afternoon/Evening/Night). Source: github.com/bhaveshchhimwal/airline-delay-analytics

Column	Type	Description
year / month / day	INT	Flight date components
origin_airport / dest_airport	VARCHAR(10)	IATA airport codes
departure_delay / arrival_delay	FLOAT	Minutes (negative = early)
is_delayed	INT	Binary: 1 if arrival_delay > 0
delay_status	VARCHAR(50)	Early/Slight/Moderate/Major
time_period	VARCHAR(20)	Morning/Afternoon/Evening/Night
distance	INT	Route distance in miles
airline	VARCHAR(100)	Full airline name